

Adam Rust

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EDUCATION

University of Wisconsin-Madison | Madison, WI

May 2027

Bachelor of Science: Computer Engineering

- GPA: 3.408 Cumulative.
- Awards: *Dean's Honor List (3 Semesters), Fluno Award.*

TECHNICAL SKILLS

Languages: SystemVerilog (HDL), C/C++, Java, Python, Bash, x86-64 Assembly, HTML/CSS.

Tools: QuestaSim, Quartus, MATLAB, LTspice, Make, Git/GitHub, Altium Designer, FreeRTOS.

Hardware & Lab: PCB Design, Mixed-Signal Routing, Oscilloscope, Digital Multimeter, Soldering.

IT : Windows OS, macOS, Linux, Hardware/Software Troubleshooting, Microsoft 365.

TECHNICAL PROJECTS

FPGA-Based Hardware Control

Aug 2025 – Dec 2025

- Applied both Moore and Mealy FSMs in SystemVerilog to coordinate internal control signals and sequential state transitions.
- Verified design functionality through comprehensive RTL simulation in QuestaSim prior to hardware deployment.

Microcontroller-Controlled LED PCB.

Jan 2026 – May 2026

- Engineered a dual-rail power distribution system to safely step down input power, ensuring voltage regulation and system hardware protection.
- Implemented hardware-level voltage supervisors and watchdog timers to prevent software and hardware lockups.

Standalone Custom 8-Bit Digital Calculator

May 2026 – Present

- Developed schematics and a 2-layer PCB layout using a direct-pin I/O keypad array, an ATtiny-1617 microcontroller, and a dedicated SPI 7-segment display.
- Integrated test points across the board layout to allow for real-time hardware diagnostics and signal troubleshooting.

Embedded RTOS Application

Jan 2026 – May 2026

- Developed a multi-task embedded application using FreeRTOS on an ARM Cortex-M processor, managing race conditions via queues, semaphores, and event groups.
- Implemented serial communication protocols (UART, I2C, SPI) to interface with external peripherals.
- Designed interrupt-driven I/O routines using NVIC priority configuration and ISRs utilizing volatile global variables.

INVOLVEMENT / LEADERSHIP

University of Wisconsin Marching Band | UW-Madison | Madison, WI

Aug 2023 – Present

Saxophone Rank Leader

- Recognized as Rank Leader to organize and prepare the saxophone section for high-profile field performances.
- Directed marching drills and mentored individual members on technique and field awareness.
- Built a teamwork-oriented culture by adapting my leadership style to better suit different communication and learning styles.
- Mentored incoming and current band members, establishing an inviting and empowering environment.

ADDITIONAL EXPERIENCE

Lands' End | Stevens Point, WI

May 2021 – Present

Warehouse Associate

- Troubleshoot production bottlenecks to maintain output goals/
- Cross-trained across multiple warehouse roles, demonstrating adaptability and strong communication skills.